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Review article

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Bridging materials innovations to sorption-based atmospheric water harvesting devices

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Abstract

The atmosphere contains 13,000 trillion litres of water, and it is a natural resource available anywhere. Sorption-based atmospheric water harvesting (SAWH) is capable of extracting water vapour using sorbent materials across a broad spectrum of relative humidity, opening new avenues to address water scarcity faced by two-thirds of the population of the world. Although substantial progress has been made, there is still a considerable barrier between fundamental research and real-world applications. In this Review, we provide a multiscale perspective for SAWH technologies that can fill existing knowledge gaps across multiple length scales. First, we elucidate water sorption mechanisms at the molecular level, approaches to understanding sorbent materials, and water transport phenomena. With microscopic insights, we bridge materials innovations to device realization, discuss strategies to enhance device-level sorption kinetics and heat transfer performance, and show that a multiscale design and optimization strategy can lead to a new opportunity space towards system thermodynamic limits. Finally, we provide an outlook for the technoeconomic, social and environmental impact of large-scale SAWH as a global water technology. By bridging materials to devices, we envision that this multiscale perspective can guide next-generation SAWH technologies and facilitate a broader impact on society and the environment.

Sections

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Introduction

With the United Nations' Sustainable Development Goal 6: clean water and sanitation, the desire for universal access to clean water has been recognized as fundamental to human health and socioeconomic development[1–3]. Conventional ways of clean water production, such as seawater desalination and wastewater treatment, are substantially restricted by the access to and the transportation of liquid water[1,4–8]. Alternatively, the atmosphere holds 13,000 trillion litres of water, which is an abundant natural water reservoir accessible to everyone on Earth[9–11]. Water sorption is a fundamental process by which water molecules spontaneously attach to another substance. It is a ubiquitous phenomenon in nature and important in industrial applications, such as catalysis[12–14], thermal energy storage[15–17] and air conditioning[18–20], although largely overlooked as a viable solution to deliver clean water[6,9,21,22]. Sorption-based atmospheric water harvesting (SAWH) is the capture and release of water vapour using sorbent materials via a temperature differential, and it can be powered by low-grade energy sources, such as sunlight and waste heat[6,9]. For a solar-driven SAWH device, water vapour is harvested in the sorbents during the night; in the daytime, sunlight is converted into heat and water vapour can be released from the sorbents, creating an environment of concentrated moisture to allow for condensation as a liquid. Traditional AWH approaches, that is, fog collection and dewing, either require mist in the air or consume electricity and, thus, have large geographical, climatic and/or socioeconomic limitations[6,9,23]. By contrast, SAWH devices can operate across a broad spectrum of relative humidity (RH), particularly in arid areas (RH < 40%), which promises decentralized water production and may address the global water shortage[1,9,24].

A global-scale assessment in 2021 indicated the great potential of solar-driven SAWH to provide clean water for a billion people and meet the average daily drinking water requirement of 5 litres per person[24]. Several companies are developing commercially available SAWH systems, but they are probably cost prohibitive in many regions where access to drinking water is needed most. And despite tremendous efforts of academic research, state-of-the-art SAWH devices can produce only up to hundreds of millilitres of water per day, which is far behind from meeting the projections[10,11,23,25–27]. This insufficient productivity is largely owing to the considerable barrier between materials innovations and device realization, which is associated with a few critical knowledge gaps across multiple length scales of SAWH technologies, from the molecular-scale water sorption mechanisms (~Å) to sorbent material characteristics (~nm), water transport phenomena (~μm), device optimization (~cm), systems implementation (~m) and the real-world impact at the global scale (~km) (Fig. 1).

This Review aims to bridge materials innovations and device realization by revisiting SAWH at different length scales and filling the knowledge gaps between these length scales. First, we discuss microscopic water sorption mechanisms and macroscopic water sorption properties, including the sorption isotherm, water diffusivity D_w and enthalpy of sorption h_s, which describe how much, how fast and how energetic water sorption is, respectively. To enable rational sorbent material design, however, an in-depth mechanistic understanding is still needed. We, thus, introduce metrology and molecular simulation techniques that can be used to understand microscopic sorption mechanisms, which fundamentally dictate macroscopic material properties. Next, we discuss materials innovations and design strategies, and identify the key factors in materials selection for device operation. Leveraging the insights gained from materials science, we provide an optimal device design framework to bridge materials innovations to device realization. We describe the basic principles of a SAWH device, elucidate optimization strategies for enhancing sorption kinetics and heat transfer, and show that the synergy of materials and device optimization can push device performance towards the thermodynamics limits. Although materials innovations and device realization will enable practical system developments of SAWH, its real-world impact remains to be realized. Accordingly, we identify the key challenges of SAWH systems by examining the technoeconomic feasibility, and we envision the broader impact of SAWH technologies on society and the environment.

Fundamental understanding of water sorption

Microscopic water sorption mechanism

The success of SAWH is largely attributed to the use of sorbent materials. These materials have excellent water affinity to attract water vapour and can accommodate captured water molecules. Sorbent materials can be classified into two main categories by distinct sorption mechanisms: nanoporous materials, which adsorb, and hygroscopic materials, which absorb (Fig. 2a,b). In adsorption, water molecules are

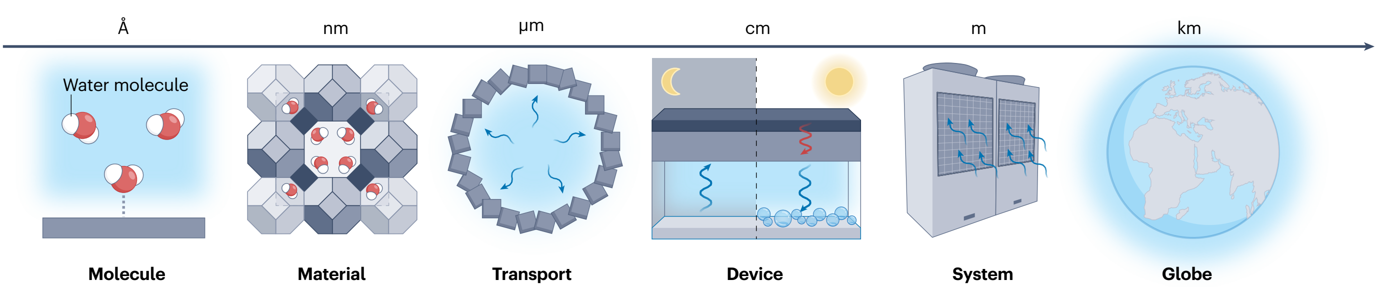

Fig. 1 | Multiscale features of SAWH. Water sorption is a fundamental process that happens at the molecular scale (~Å), which dictates sorption characteristics of materials (~nm) and water transport phenomena in the sorbents (~μm). At the device level, the optimization of sorption kinetics and heat transfer is crucial (~cm) and highly depends on the choice of sorbent materials. After being scaled up, real-world sorption-based atmospheric water harvesting (SAWH) systems may be deployed in households or commercial buildings (~m), which can lead to a worldwide impact (~km). Still, there are a few critical knowledge gaps across multiple length scales in the design of SAWH, leading to considerable barriers from materials innovations to device realization. Blue arrows indicate movement of water vapour; red arrows indicate heat flow.

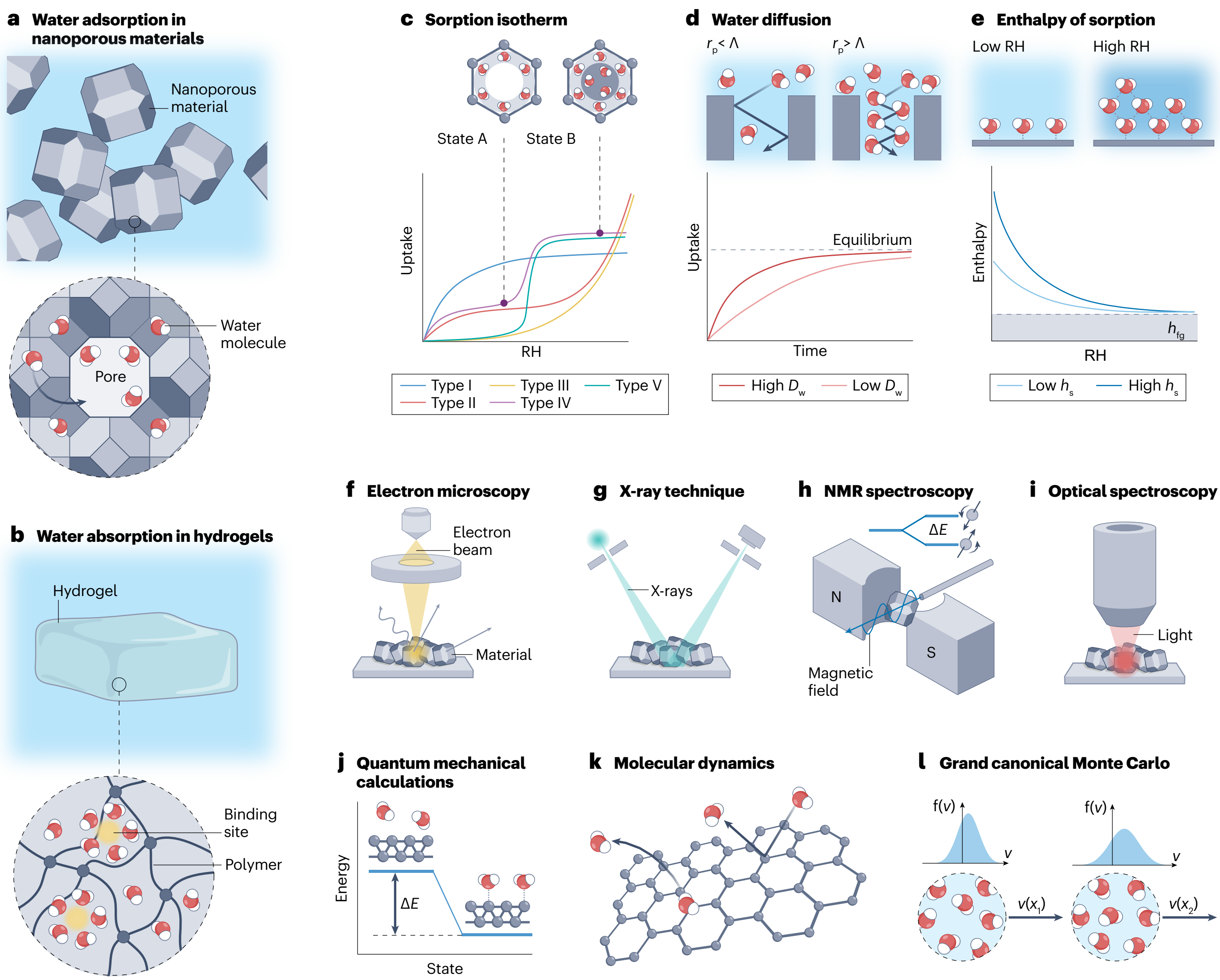

Fig. 2 | Understanding microscopic mechanisms and macroscopic properties of water sorption. **a**, Water adsorption in nanoporous materials, which initiates by binding water molecules at preferential sites and is followed by a pore filling process. **b**, Water absorption in hydrogels. Water molecules are incorporated into the bulk material and retained via intermolecular interactions. **c**, Water sorption isotherms, isothermal gravimetric water uptake as a function of relative humidity (RH). The schematics illustrate the filling of water molecules in the sorbent at two levels of RH. **d**, Water uptake as a function of time during sorption, which is dictated by water diffusivity D_w. The schematics show distinct kinetic behaviours when the pore size r_p and the mean free path Λ of water molecules are different, which affects water diffusivity. **e**, Enthalpy of sorption h_s as a function of RH, which generally saturates to the heat of vaporization of water h_{fg} at high RH. The insets show the configurations of captured water molecules at low RH and high RH. **f**, Electron microscopy resolves real-space information of materials, including morphology and feature size. **g**, X-ray techniques provide high-resolution structure information of materials. **h**, NMR spectroscopy probes perturbations in the local structure and the stability of sorbent materials by detecting spin properties of atomic nuclei in magnetic fields. **i**, Optical spectroscopy reveals chemical fingerprints of molecules or materials, such as information of bonding and chemical constituents. **j**, Quantum mechanical calculations enable fundamental understanding of binding energy and thermodynamic states. **k**, Molecular dynamics simulations model the transient states of sorption systems and resolve both equilibrium and dynamic properties. **l**, Grand canonical Monte Carlo simulations statistically sample thermodynamic ensembles of sorption.

attached to the pore surface, whereas in absorption, water molecules enter into the volume of the material. The fundamental differences in the material chemistry and structure between nanoporous materials and hygroscopic materials dictate their distinct microscopic water sorption mechanisms.

The family of nanoporous materials includes crystalline metal–organic frameworks (MOFs), covalent–organic frameworks (COFs) and zeolites, as well as amorphous silica gels. Adsorption initiates by binding water molecules on the preferential binding sites and is followed by the growth of water networks via a pore-filling process

(Fig. 2a, bottom). The complex structure of water molecules confined in the pore is sometimes referred to as water clusters, which is different from layer-by-layer adsorption because of the nanoconfinement effect[28–31]. The microscopic pore features, including pore diameter, pore volume and surface hydrophilicity, are the most critical factors that affect macroscopic water sorption properties in nanoporous materials. Typically, a larger pore size and higher hydrophilicity facilitate the accommodation of more water molecules, whereas smaller pore size (<1 nm) enables adsorption at low relative humidity (RH < 30%). Theoretical and experimental studies found that for sorbents with a pore size above the critical diameter (≈2 nm), capillary condensation of water occurs at room temperature[31–34]. As a thermodynamically irreversible process, however, capillary condensation induces a hysteresis between adsorption and desorption processes, which is unfavourable for SAWH. Therefore, it is desirable for a nanoporous material sorbent to have an appropriate pore size (<2 nm) and highly hydrophilic pore surfaces. Silica gel has been commonly used as a desiccant, but it may not be desirable for SAWH applications that require a sorbent to possess high water uptake, sorption at low RH, low energy consumption to release captured water, and the ability to tailor material properties[9,35–38]. Therefore, we focus our discussion on more representative crystalline nanoporous materials, that is, MOFs and zeolites, which are relatively well-investigated and more desirable for SAWH applications.

Hygroscopic materials, including hydrogels, salts and their composites, undergo absorption when exposed to a humid environment (Fig. 2b). The absorption characteristics of water in different hygroscopic materials may vary. In hydrogels, water molecules diffuse into the bulk of the material as driven by an osmotic pressure difference and are retained in bulk via the interactions with the ionic binding sites or intermolecular hydrogen bonding. For porous hydrogels, vapour diffuses through the interconnected gaseous micropores and permeates across the surfaces of the pores into the polymer network[39]. The features of hydrogels are considerably larger than those of nanoporous materials by one to several orders of magnitude[39–41]. The backbones of hydrogels are also flexible compared with nanoporous materials, so they can accommodate more absorbed water molecules via swelling[39,42–44]. Pure hydrogels usually capture vapour at only high RH owing to comparably low hygroscopicity and undergo a slow diffusion process[42,45,46]. Salts, by contrast, have high hygroscopicity and absorb water vapour through hydration and deliquescence[26,30,47,48]. Although salts have a high water uptake at low RH, their solid structures cannot be retained owing to deliquescence. This can cause agglomeration, leaching of salts, reduction of vapour permeability, and corrosion, which lead to marked degradation with cycling operation[26,30,47,48]. To overcome these limitations, pure hydrogels can be used as host matrices to impregnate salts. The resulting composite materials, or hygroscopic hydrogels, can prevent the salts from leaching while taking advantage of their high hygroscopicity, which has merited tremendous attention for SAWH[36,38,40,42,49,50].

Macroscopic water sorption properties

Macroscopic water sorption properties are key to SAWH technologies. The sorption isotherm shows the equilibrium water uptake of the sorbent material as a function of RH at a certain temperature (T). The water harvesting capacity is determined by the difference in the equilibrium water uptakes at sorption and desorption temperatures[10,37,51]. According to the International Union of Pure and Applied Chemistry (IUPAC), there are six well-defined types of sorption isotherms[52] (Fig. 2c). Although being capable of capturing water at low RH, a sorbent with a type I isotherm can be difficult to regenerate and is less ideal. Sorbents with type II and III isotherms can harvest a substantial amount of water only at high RH. It is commonly perceived that the step-shaped isotherms, types IV and V, are favoured[51,53,54]. The key characteristics of type IV and V sorption isotherms are the maximum water uptake (q_{max}), wherein water uptake saturates (Fig. 2c, state B), and the inflection RH point (RH_I), wherein the step occurs. The type VI isotherm is rarely seen in SAWH applications, and therefore, it is not shown in Fig. 2c and is beyond the scope of our current discussion. At different levels of RH (Fig. 2c, states A and B), water molecules may form various structures inside the nanopores owing to complex molecular interactions, leading to an inflection in the isotherm. Type IV and V isotherms can be commonly found in nanoporous sorbent materials, and type II and III isotherms are characteristic of hygroscopic materials[17,55]. The most standard way of measuring an isotherm is the thermogravimetric approach using a sorption analyser, which measures sorption in a vapour–nitrogen environment or a pure vapour environment. The sorption isotherm can be measured with recorded sorbent mass change over a series of RH steps (typically from 0% to 95%).

Water diffusivity D_w, which describes how fast water, in the form of liquid or vapour, can be captured and released, is critical to water productivity. D_w can be the intracrystalline diffusivity of water in loosely packed nanoporous materials or bulk diffusivity of water in hydrogels. Using the thermogravimetric approach, the water uptake is measured as a function of time, through which D_w can be obtained by fitting to Fick's law of diffusion[56,57] (Fig. 2d). D_w can be strongly affected by the characteristic feature size of materials, for example, the pore size r_p of nanoporous sorbent or the mesh size of hydrogels relative to the mean free path Λ and the size r_m of water molecules[58,59]. We note that D_w, as an intrinsic property, should be used to depict water sorption kinetics rather than a time constant of the sorption process, which may have strong dependence on the sample size[10,37,57].

The enthalpy of sorption h_s is another fundamental sorption property that bridges the molecular binding energy and SAWH device energy consumption. Water molecules can be bonded to the sorbents and form one or multiple layers by physisorption, including van der Waals forces or hydrogen bonds (H-bonds, $h_s \approx 44–60$ kJ mol^{-1}), or chemisorption owing to the formation of chemical bonds ($h_s \approx 100$ kJ mol^{-1})[30,31,53,60,61]. With the continuous filling of water molecules, H-bonds gradually dominate intermolecular interactions, so h_s approaches the heat of vaporization of water ($h_{fg} = 44$ kJ mol^{-1} at room temperature) at high RH[62] (Fig. 2e). Derived from the Clausius–Clapeyron relation as a temperature-average property, the isosteric h_s is typically used to evaluate a sorbent[31,60,62,63]. Alternatively, h_s can be characterized as a function of water uptake by measuring the change of mass and heat flow over time using thermogravimetric analysis and differential scanning calorimetry measurements, respectively[62]. The results indicate a general trend that h_s decreases with RH towards h_{fg}, which is consistent with the microscopic understanding of sorption mechanisms[28,62].

According to the above qualitative descriptions, it is clear that macroscopic water sorption properties are determined by microscopic water sorption mechanisms. However, the design of sorbent materials still heavily relies on physical intuition and trial-and-error-based approaches owing to the lack of complete quantitative understanding to bridge the knowledge gap. Statistical mechanical models have been developed in the past with inputs from experimental data or simulations, providing an analytical yet simplified view to understand water sorption[64–67]. More recently, integrated efforts of advanced metrology and molecular simulations have aided in developing a strong mechanistic understanding to fill the knowledge gap, which promises to improve sorbent material performance.

Metrology to understand water sorption

To understand water sorption, a complete set of physical and chemical information is essential, such as material structures, preferential binding sites, and the type and strength of bonds. The physical and chemical information about materials and molecular interactions can be accessed by advanced metrology using different sources of excitation. Electron microscopy has been the primary tool to directly probe visual information about materials, such as structure, morphology and feature size[68–70] (Fig. 2f). X-ray techniques have been instrumental in providing high-resolution structural information about sorbent materials, such as material lattice parameters and the coordination of atoms, which serve as direct evidence of how water molecules interact with sorbents[28,29,71,72] (Fig. 2g). NMR spectroscopy analyses chemical structure of materials using isotopes and is capable of elucidating water-sorbent interactions[31,73–75] (Fig. 2h). Aside from the techniques providing structural information, optical spectroscopy, including infrared (IR) spectroscopy and Raman spectroscopy, reveals chemical fingerprints of molecules and materials[40,63,72,76–80] (Fig. 2i). Owing to its high sensitivity to detect atom vibrations and polar interactions of water molecules, optical spectroscopy enables monitoring the behaviour of water molecules. Table 1 summarizes these metrology techniques and their applications. Here, we highlight a few techniques that have advanced the fundamental understanding of microscopic sorption, which can be further translated into material design.

Single-crystal X-ray diffraction (SCXRD) has been instrumental for water sorption studies[28,29,81]. It was applied to investigate molecular-level structural changes of MOFs and the evolution of water structures during sorption[28,29]. By following the progression of maximum water density spreading away from the linkers and ultimately centering in the nanopore, in situ SCXRD provided insights into the origin of step-shaped isotherms[29]. These findings successfully revealed the preferential binding sites of water molecules and enabled the interpretation of water sorption in three stages – seeding, clustering and networking – corresponding to the three regimes of the step-shaped isotherm[28,29]. Transmission electron microscopy (TEM) enables direct visualization of the atomic structure of sorbent materials that are less than several hundreds of nanometres thick[82–84]. However, it is still challenging for conventional TEM to visualize how an individual or a group of water molecules interact with the pore environment. As a possible alternative, in situ scanning TEM (STEM) has successfully directly visualized binding sites, orientation and structural changes during sorption of small molecules in zeolites[85–87], which has not been demonstrated yet for water sorption. These enhanced capabilities could potentially facilitate the observation and understanding of how water sorption evolves over time at the molecular level. With the development of advanced metrology to characterize the physical structure of materials and molecules, including in situ SCXRD[28,29,81] and STEM[85–87], we envision that a new avenue is being paved to study water-sorbent interactions and to tackle unresolved questions at the molecular scale.

For NMR measurements, ^{1}H-NMR and ^{13}C-NMR have proven useful in tracking chemical structural changes of sorbent materials at different water uptakes and elucidating the preferential adsorption sites and the degradation pathways[73,88,89]. Many metals used to construct MOFs have isotopes, such as ^{27}Al, that are NMR active. These isotopes allow the measurement of changes in the local environment of the metal clusters during sorption, revealing the chemical dynamics of water sorbent interactions[31,88]. NMR is also able to provide a plethora of physical and chemical information of hydrogels, including mesh size, crosslinking density and structure changes during swelling, which provides a scope to detect vapour diffusion[74,90]. Optical spectroscopy has been implemented to detect interactions between water molecules and sorbents, and among water molecules in the H-bond networks in nanoporous materials[72,76,77,79,91,92] and hydrogels[40,42,71,93]. For example, IR spectroscopy was used to characterize the effect of anion exchange on the O–H bond stretching, which showed that the surface chemistry substantially affects sorption of water molecules on the preferential sites[77]. By combining in situ Raman spectroscopy with advanced imaging techniques, heterogeneities in the local chemical environment, such as defects and morphological characteristics, were shown to have a strong impact on the host–guest interactions[80]. More interestingly, this approach further quantified water uptake at the single-crystal level, which revealed the single-crystal limit of the water harvesting capabilities of MOFs[78]. Owing to their high selectivity in detecting water molecules, NMR and IR spectroscopies have also been applied to track their motion, providing microscopic insights into water diffusion[75,94–96].

Table 1 | Metrology techniques to understand the fundamental water sorption mechanism

Metrology		Applications	
X-ray techniques		Structural information of materials and water molecules	
NMR spectroscopy		Chemical structure, specific arrangements of atoms, dynamics of the molecules and perturbations in the local structure	
Electron microscopy	SEM	Imaging of material morphology and crystal size	
	TEM	Imaging of material structure and interfaces, and visualization of the evolution of molecular sorption	
Optical spectroscopy	IR spectroscopy	Exotic states of water molecules in H-bond networks	Strong dipole interactions
	Raman spectroscopy		Polarizability change by molecular vibrations

IR, infrared; SEM, scanning electron microscopy; TEM, transmission electron microscopy.

Molecular simulations of water sorption

As a complement to experimental metrology, molecular simulation studies idealized sorbent systems to enhance the understanding of water sorption. By gaining insights beyond the spatial and temporal detection limit of metrology, molecular simulations can bridge the gap between microscopic sorption mechanisms and macroscopic material properties. However, the critical hurdle of modelling water sorption relates to how much information a simulation can provide at each time and length scale. The difficulty typically lies in achieving computational efficiency and accuracy simultaneously, which is largely hindered by incomplete knowledge of energetic interactions between atoms in these complex systems. There are three primary molecular simulation techniques that study water sorption at different theoretical levels, from microscopic to mesoscopic scales, including quantum mechanical calculations, molecular dynamics (MD) simulations and grand canonical Monte Carlo (GCMC) simulations[55,97]. Table 2 summarizes these techniques and their applicable time and length scales.

Table 2 | Molecular simulation techniques to understand the fundamental water sorption mechanism

Molecular simulations		Applications	Timescale	Length scale
Quantum mechanical calculations		Binding sites and energy, thermodynamic states	–	~Å – ~nm
Molecular dynamics simulations	Ab initio	Equilibrium and dynamical properties of water sorption	~fs – ~ps	~Å – ~nm
	Classical		~ps – ~ns	~nm
Grand canonical Monte Carlo simulations		Efficient water sorption modelling and isotherm simulation	–	~nm

Quantum mechanical calculations, particularly density functional theory (DFT) methods, are instrumental in predicting the binding energy on the pore surface and thermodynamic states of water molecules in the pore environment (on the angstrom length scale)[55,97,98] (Fig. 2j). Enabled by DFT calculations, researchers can theoretically detect the binding sites of water molecules in MOFs[28,99], zeolites[100] and hydrogels[43]. These studies precisely probe the energetic states of water molecules and elucidate how surface chemistry affects the preferential sorption site at open metal sites[99] or polymer chains[43]. Although quantum mechanical calculations provide accurate static descriptions of water sorption, the dynamics of sorption are too expensive to compute[97]. Still, quantum mechanical methods enable first-principles evaluation of force fields, which can be implemented in MD and GCMC simulations, providing a more accurate description of energetic interactions between atoms compared with the use of empirical force fields[101–103].

MD simulations, by contrast, can predict both equilibrium and dynamic properties of a larger system at finite temperature based on a predefined force field (on the nanometre length scale and the nanosecond timescale)[55,97]. By solving the Newtonian equations of motion of atoms, the evolution of water molecules can be traced such that not only equilibrium states of water uptake and energy (Fig. 2k) but also dynamic properties, such as water diffusivity, can be calculated[97,104]. Classical MD simulations have been applied to visualize the spatial orientation and distribution of water molecules[92,105], investigate water motion in the time domain and frequency domain[76,106,107], and compute self-diffusion coefficients using Einstein's relation[107]. MD simulations revealed that water molecules within a nanopore are distributed across three distinct spatial sectors, and that the water diffusivity in each sector differs[92]. As the distance from the pore surface increases, the dynamics of water molecules, characterized by water diffusivity, become closer to liquid water, which is consistent with the microscopic understanding of sorption mechanisms[28,62,92]. A similar conclusion was reached based on the observation of three different modes of intermolecular motion in water molecules, each characterized by distinct frequencies[107].

Nevertheless, the key limitation of MD simulations is the uncertainty of the force fields at each timestep because of the polar nature of water molecules and their energetic interactions with each other and the sorbent. Ab initio MD (AIMD) simulations can incorporate electronic polarization effects of water molecules and sorbent materials and provide unbiased forces at each modelling timestep by performing DFT calculations[55,76,106]. However, owing to the higher computational cost, the timescale accessible for AIMD simulations is more limited (on the picosecond timescale) than those of classical MD simulations[55].

Different from MD simulations that study a system in a time-dependent manner, GCMC simulations statistically sample the thermodynamic ensembles of a system (on the nanometre length scale, Fig. 2l), which investigates water sorption more efficiently with reasonable accuracy[55,97,108,109]. Taking material structures and a force field as inputs, GCMC simulations have widely served as a powerful tool to model water sorption isotherms[108,109]. GCMC simulations have been extensively applied to simulate the continuous water sorption processes with increasing RHs, such as in nanoporous materials including MIL-101(Cr)[105], KMF-1 (ref. 110) and MOF-801 (ref. 111). More importantly, owing to the relatively high computational efficiency, GCMC simulations can predict water sorption isotherms based on material design strategies, providing a clear physical picture from microscopic sorption mechanisms to engineer sorbent materials[89,97,110,111].

Materials characteristics and innovations

The physical insights gained from advanced metrology and molecular simulations enable the development of novel sorbent materials by engineering water sorption properties. In this section, we first identify key sorbent characteristics in six dimensions that ultimately determine the performance of sorbent materials, as required by practical device operation. Then, we provide a few examples to show how the microscopic understanding of sorption mechanisms can guide the design of sorbent materials with tunable macroscopic water sorption properties.

Metrics to quantify sorbent performance

Figure 3a shows an SEM image of a MIL-101(Cr) crystal with an octahedral morphology[83]. Its nanoscale crystalline features can be resolved under the TEM[83] (Fig. 3b). Figure 3c shows features of porous hydrogels, including the random porous skeleton, pore structures and surface wall structures at different length scales, which are all considerably larger than those of nanoporous materials[40]. These differences in microscopic features between nanoporous materials and hydrogels result in their distinct macroscopic sorption properties, especially their isotherms. Figure 3d shows the water sorption isotherms of several state-of-the-art nanoporous materials and pure hydrogels characterized by sorption analysers[9,23,47,77,112–115], and Fig. 3e shows the isotherms of two salts and several hygroscopic composite materials[25,26,38,40,47,116,117]. In SAWH applications, nanoporous materials typically have an isotherm of type IV or V. The RH_i is usually lower than 40%, and the water uptake of the materials generally saturates to the q_{max} at 60% RH. Pure hydrogels behave with a type II or III isotherm and can only harvest substantial moisture above 60% RH, whereas hygroscopic hydrogels show considerably higher water uptake at RH well below 60% owing to the impregnation of salts.

In addition to exhibiting a high water uptake with negligible hysteresis, fast water sorption and low h_s, an ideal sorbent is also expected to be air stable and water stable and maintain its capacity after repeated cycles of sorption–desorption, as well as scalable and low cost. Figure 3f compares three promising material candidates for next-generation SAWH – MOFs, zeolites and hydrogels – in six key dimensions: uptake, kinetics, enthalpy, scalability, stability and cost, by relative scores from one (the worst) to three (the best). These material characteristics are important for device realization and system development to enable real-world impact. Quantitative evaluations of uptake, kinetics and enthalpy are derived respectively from literature data of gravimetric uptake, water diffusivity and enthalpy compiled in Table 3, whereas scalability and stability are assessed qualitatively. The evaluation of cost is conducted by examining the market price, with further details available in the section 'Technoeconomic feasibility of SAWH'. Note that although D_w, an intrinsic material property, shows considerable variation among different materials, it is critical to account for the specific sizes of materials when comparing their sorption kinetics because sorption kinetics are determined by both water diffusivity and the size of sorbents. Moreover, a higher score of enthalpy indicates a lower h_s, that is, desorption is more energy efficient; a higher score of cost means the cost of materials is relatively lower. Among the three categories of materials, hydrogels show high water uptake but are currently limited by slow water sorption[42,45,46,61]. By contrast, zeolites possess fast sorption kinetics, whereas their applications are hindered by relatively low water uptake and high h_s (refs. 37,61,62). MOFs generally exhibit promising potential in terms of a low enthalpy and relatively fast kinetics, but their cost and scalability scores are lower than those of the other materials and are a priority to improve[6,30,61,118–121]. In addition to the material characteristics influencing score, Table 3 summarizes a few device parameters of MOFs and COFs, zeolites, and hydrogels, which can serve as guidelines for material selection for device design. Notably, the sorbent material density is an important design parameter, which determines the volumetric water uptake[23]. Highly packed sorbents possess a higher volumetric uptake and can increase device compactness.

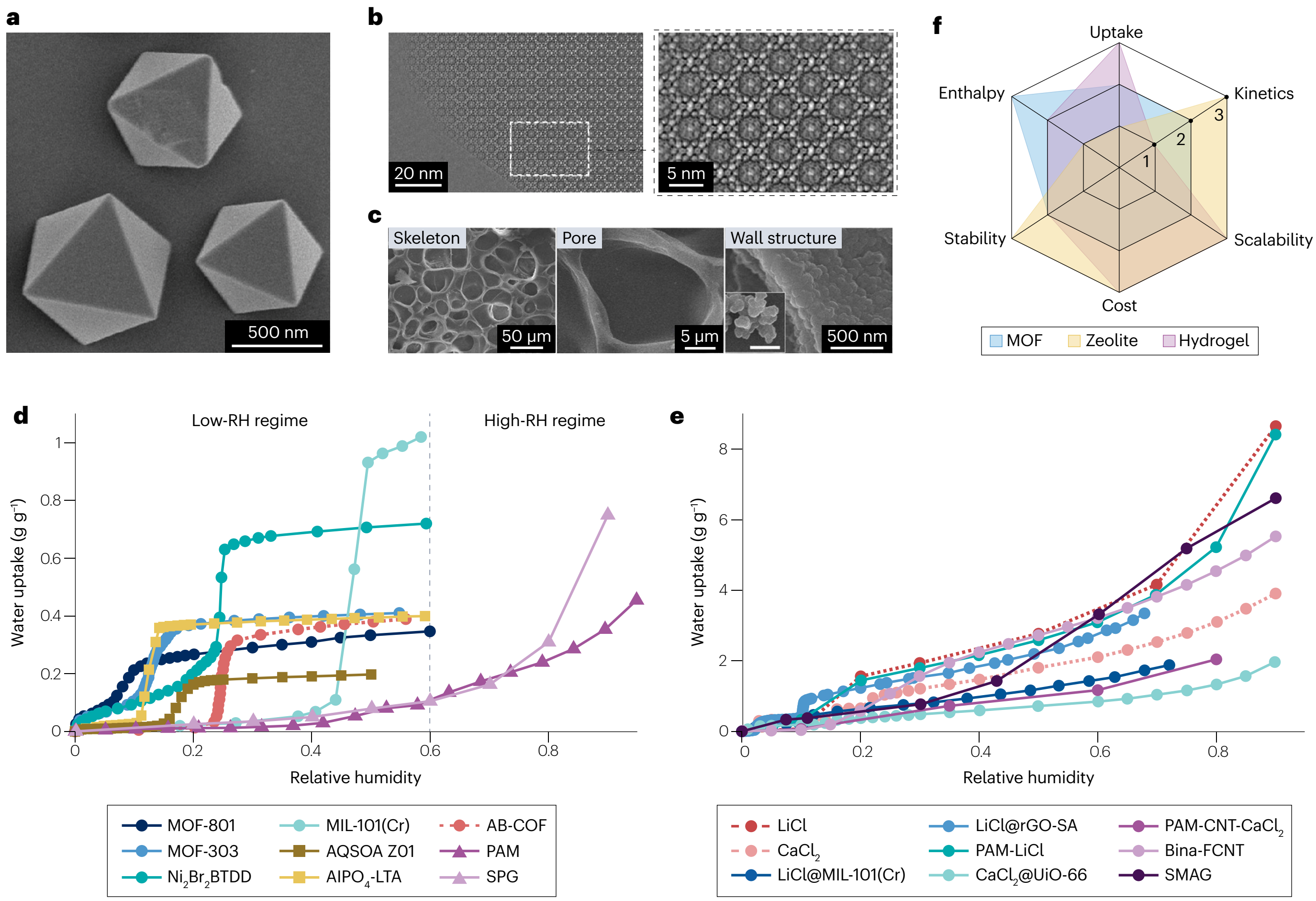

Fig. 3 | Materials innovations and characteristics for water sorption. **a**, Scanning electron microscopy (SEM) image of a MIL-101 crystal with an octahedral morphology[83]. **b**, High-resolution transmission electron microscopy (TEM) images of MIL-101 (left panel) and the corresponding magnified image of the highlighted area (right panel), showing an array of mesoporous cages with well-resolved cage surface features[83]. **c**, SEM images of the super moisture-absorbent gel, showing the polymer skeleton, micrometre-sized pore and fine features of wall structures in different magnifications (from left to right)[40]. **d**, Water sorption isotherms of representative sorbent materials[9,23,47,77,112–115]. The shape of isotherms is dictated by fundamental differences in sorption mechanisms. Nanoporous material sorbents typically have a step-shaped isotherm and harvest water vapour in the low-relative humidity (RH) regime (<60% RH), whereas pure hydrogels with a comparably low hygroscopicity capture moisture in the high-RH regime (>60% RH). **e**, Water sorption isotherms of salts and hygroscopic composite materials[25,26,38,40,47,116,117]. Because of the substantially enhanced hygroscopicity, composite materials can substantially uptake water at RH well below 60%. **f**, Comparison of material characteristics of the three common sorbent materials in six dimensions: uptake, kinetics, enthalpy, scalability, stability and cost. A higher value of enthalpy indicates a lower enthalpy of sorption. A higher score of cost means the cost of materials is relatively lower. COF, covalent–organic frameworks; CNT, carbon nanotube; MOF, metal–organic framework; PAM, polyacrylamide. Panels **a** and **b** reprinted with permission from ref. 83, American Chemical Society. Panel **c** adapted with permission from ref. 40, Wiley.

Table 3 | Key material properties for SAWH device design

Material	Isotherm type	Density ρ ($g_{sorbent}\,cm^{-3}$)	Gravimetric maximum uptake[a] q_{max} ($g_{water}\,g_{sorbent}^{-1}$)	Volumetric maximum uptake[a] $q_{max,v}$ ($g_{water}\,cm^{-3}$)	Water diffusivity[b] D_w ($m^2\,s^{-1}$)	Sorption enthalpy h_s ($kJ\,mol^{-1}$)	Operating RH RH_{op} (%)[c]	Desorption temperature[c] T_{op} (°C)
MOF or COF	I, IV, V	0.7–1.4	0.32–1.05	0.36–0.72	1×10^{-18}–1×10^{-16}	45–60	15–60	50–80
Zeolite	I, IV, V	1.4–1.5	0.25–0.54	0.42–0.54	1×10^{-16}–1×10^{-10}	55–75	0–30	75–120
Hydrogel	II, III	1.19–2.15	0.15–3.4	0.15–3.1	1×10^{-10} (liquid)–1×10^{-6} (vapour)	44–80	30–60	40–80

COF, covalent–organic framework; MOF, metal–organic framework; RH, relative humidity; SAWH, sorption-based atmospheric water harvesting. [a]The water uptakes are reported at 60% RH. [b]Water diffusivity is reported at room temperature in the unit of square metres per second ($m^2\,s^{-1}$) for consistency. The range for hydrogels is set by the fundamental water diffusion limits, with the lower limit being pure liquid diffusion and the upper limit being pure gas diffusion[58,59]. Notably, water diffusivity is influenced by the characteristic feature size of sorbents relative to the mean free path and the size of water molecules[58,59] (Fig. 3b,c). Although water diffusivity varies substantially across different materials, it is essential to consider their finite sizes when comparing sorption kinetics. Additionally, the effective water diffusivity, which accounts for both liquid and vapour water diffusion, is proposed to describe water transport in the hydrogel[39]. The effective water diffusivity is calculated by multiplying the water diffusivity by the density of either water liquid or water vapour, yielding a unit of kilogram per metre per second ($kg\,m^{-1}\,s^{-1}$), which can elucidate the interplay between the liquid and vapour transport[39]. [c]For sorbents with a step-shaped isotherm, the operating RH and temperature can be RH_I and regeneration temperature T_{reg}, respectively, which is the minimum desorption temperature required to lower the RH in the sorbent such that the equilibrium water uptake is below the isotherm step[37]. T_{reg} depends on the temperature of the condenser T_{cond} and RH_I, and it needs to satisfy $\frac{p_{sat}(T_{cond})}{p_{sat}(T_{reg})} < RH_I$ (ref. 37). For sorbents without a step in the isotherm, such as hydrogels, the operating RH and temperature can be the practical optimal conditions to release most of the water[40,42,43,45,117].

Material design strategies

One representative approach to designing macroscopic sorption properties by changing the microscopic features is known as the multivariate strategy, which takes advantage of the reticular nature of nanoporous materials[122]. By using different types of chemical constituents, such as organic linkers, the multivariate strategy enables precise modulation of the binding strength of initial water sorption sites, as well as active control over the shape of the resulting water sorption isotherms[28,31,77,123,124]. For example, the onset of pore-filling was modulated in the Ni_2X_2BTDD (X = OH, F, Cl, Br) series by using anion exchange to achieve a record water uptake (1.07 $g_{water}\,g_{sorbent}^{-1}$)[77]. Specifically, replacing Cl^- with the larger yet less hydrophilic Br^- results in a substantial reduction of inflection RH point, whereas the insertion of more hydrophilic F^- and OH^- does not facilitate pore-filling at lower relative humidity because the thermodynamic favorability of water pore-filling depends more on the pore diameter than on the water affinity to pore surfaces[77]. In addition, h_s could be reduced from 54 to 49 kJ mol^{-1} by modulating the ratio of two organic linkers in the multivariate PT-MOF series (formulated as $[Al(OH)\,(PZDC)_{1-x}(TDC)_x]$)[123]. It was observed that the surface chemistry of linkers influences preferential water binding and the formation of H-bonding, and a linker with lower hydrophilicity can effectively reduce h_s without sacrificing the water uptake[28,123].

Synthetically combining materials into composites is a general strategy to tune sorption isotherms. It has proven to be especially effective in the case of hygroscopic hydrogels, which combine the exceptional hygroscopicity of salts with the large surface areas and water retention of their host matrices[26,38,42,117]. However, the leaching of salts owing to deliquescence at high RH, resulting in the loss of sorbent and corrosion, remains a challenge, particularly at desirable high salt contents[48,60]. The strategy has also been adopted for other hygroscopic components and host matrices, such as salts in MOFs[47,125].

Although notable progress has been made in engineering the sorption isotherm and h_s, more efforts should be devoted to tuning the D_w of sorbent materials. General guidelines for using the multivariate strategy to tune the water diffusivity of nanoporous materials are still lacking, owing to limited mechanistic understanding of water molecule transport in nanopores. In hydrogels, because water sorption is coupled with mechanical deformation, the design of the cross-linking density and polymer mesh size can control both water absorption and polymer network expansion, creating engineering space for optimizing swelling kinetics and sorption properties[38,39,42–44]. The D_w can also be tuned through modifications of polymer chemical composition[42,43] and pore structures[40,45,126].

Device design and optimization

The design of SAWH devices relies on the macroscopic sorption properties mentioned in the previous section (sorption isotherm, D_w and h_s) as inputs. However, these material properties are not the only parameters that determine device performance. Equally important are device-level design parameters, which govern the interaction between sorbent materials and other components of the device, such as the solar absorber and the condenser. During sorption, the macroscopic sorption properties indeed play the predominant part because it is always desirable to maximize the water uptake and sorption kinetics (Table 3). During desorption, however, device-level parameters can be more important because heat must be utilized to reach an appropriate desorption temperature and overcome the energy barrier h_s (Table 3). The water production is, thus, the result of both material properties and device parameters, which together require a multiscale design and optimization strategy via the manipulation of transport processes, that is, sorption kinetics and heat transfer. We generalize a multiscale device framework with macroscopic water sorption properties as inputs, and address the large opportunity space for device design and optimization.

Device working principles and performance metrics

SAWH devices can either operate in a passive daily cycle powered by sunlight or operate with multiple cycles per day with auxiliary device components for heating, cooling and air supply, which can be powered by photovoltaic (PV) cells[112]. A daily cycle device is designed to be regenerated passively by sunlight and typically consists of a transparent cover, a solar absorber, a packed sorbent layer and a condenser (Fig. 4a). Air gaps exist between both the cover and the solar absorber, and the sorbent layer and the condenser. During the night, the sorbent layer is exposed to the ambient. Water vapour molecules diffuse into the sorbent layer and are captured by sorbent materials. During the day, sunlight is converted to heat by the solar absorber. The air gap between the transparent convection cover and the solar absorber prevents convective heat loss to the environment. Heat from the solar absorber is conducted to the sorbent layer and desorbs harvested water. To effectively conduct heat and enable desorption, nanoporous sorbent materials are typically packed in a porous thermal binder, such as a copper foam, in the sorbent layer. After desorption, water vapour diffuses through the air gap between the sorbent layer and the condenser. Although we mainly focus on a representative solar-powered device operating in a daily cycle as its fully passive nature has attracted extensive research interest, the design principle can also be generalized to other types of SAWH devices.

To evaluate the performance of SAWH devices, the most relevant figure of merit is the energy-to-water thermal efficiency η_t (other variables are listed in Box 1):

$$\eta_t = \frac{m_w h_{fg}}{Q}, \quad (1)$$

where m_w is the mass of collected water, h_{fg} is the water vaporization enthalpy, and Q is total amount of energy input in a device operation cycle, that is, daily solar irradiance for a solar-powered device. η_t measures the efficiency of energy-to-water conversion in a practical manner, which accounts for the effectiveness of thermal energy conversion for regeneration, vapour condensation and water collection. Meanwhile, the specific water productivity (SWP, l m^{-2} day^{-1}) is used to quantify device performance as the water production per unit solar absorbing area per day. When the footprint of the device and daily water harvesting rate become critical design factors, SWP becomes a more important parameter. We introduce the third performance metric, specific energy consumption (SEC, kWh l^{-1}), which is defined by the ratio of energy input by the amount of water harvested. The SEC is a commonly used metric to describe water-related technologies, such as heat pumps[127,128] and desalination[4,129]. While being inversely proportional to η_t, it offers practical guidance by relating the amount of water production to the amount of energy consumption.

Optimizing water sorption kinetics

To demonstrate that the optimization of water sorption kinetics is a critical step to bridge materials and devices, our discussion mainly focuses on nanoporous materials that are relatively well-understood. We formulate the sorption kinetic equations using the two-concentration model (TCM), which couples the vapour concentrations in the environment (C) and within the sorbent material (C_μ)[9,37,130]. The TCM considers water vapour transport across multiple length scales and serves as a key tool to bridge materials and devices[9,37,130]. Next, we discuss the opportunity space enabled by engineering vapour diffusion in the sorbent layer with a packing porosity ε, which is the ratio of the void volume to the total

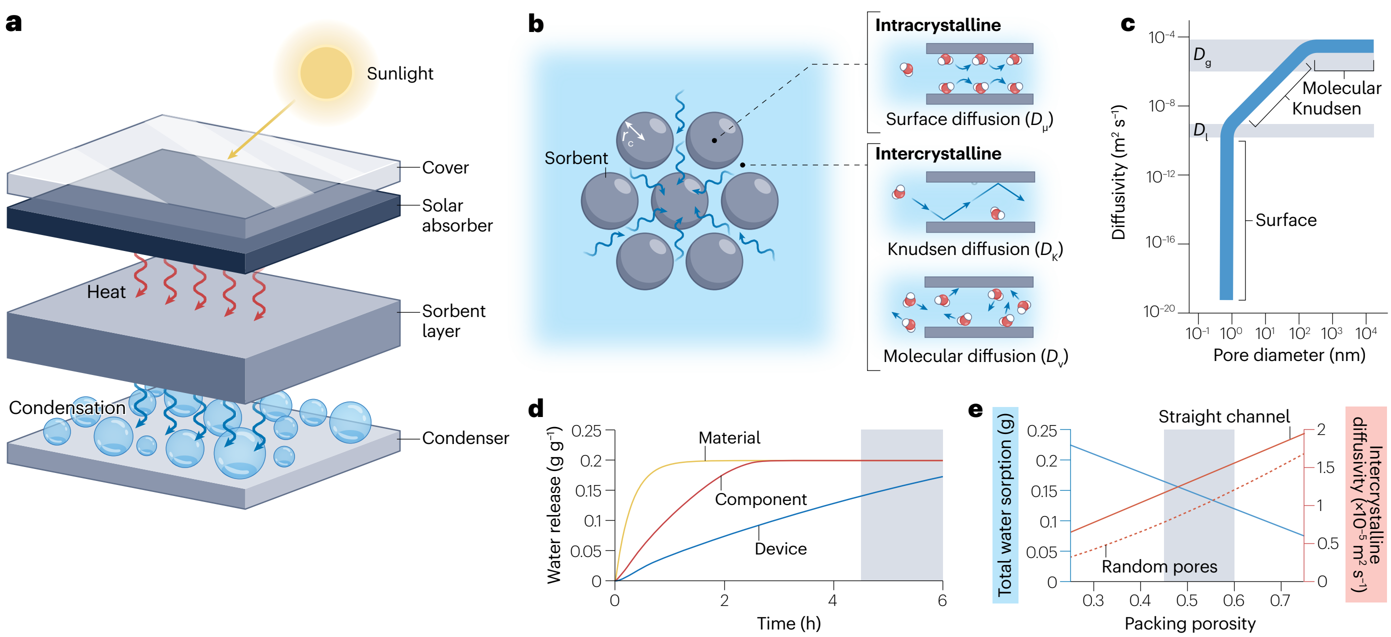

Fig. 4 | SAWH device modelling and its optimization strategy. a, A sorption-based atmospheric water harvesting (SAWH) device consisting of a transparent cover, a solar absorber, a sorbent layer and a condenser. Sunlight is transmitted through a transparent cover and absorbed by a solar absorber, which is converted to heat for desorption. Desorbed vapour condenses as liquid water on a cold condenser surface. **b**, Diffusion regimes in the sorbent layer at different length scales. The intracrystalline diffusion, that is, surface diffusion, happens within the material as the pore size r_p is similar to the size of the water molecule ($r_m \approx 3$ Å). The intercrystalline diffusion includes both Knudsen diffusion wherein the spacing between the crystals is close to the mean free path of water molecules Λ, that is, $r_p \approx \Lambda$, and molecular diffusion in the micropores of hydrogels or the porous sorbent layer, where $r_p \gg \Lambda$. **c**, Vapour diffusivity as a function of pore size[58,59]. The diffusivity orders of magnitude show the different diffusion regimes at different length scales, that is, surface diffusion ($r_p \approx r_m$), Knudsen diffusion ($r_p \approx \Lambda$) and molecular diffusion ($r_p \gg \Lambda$). D_g and D_l are the diffusivity of water in the gas phase and liquid phase, respectively. **d**, Example evolution of water release from the material (sorbent), component (the sorbent layer) and the device during desorption. The large differences of the three curves are owing to kinetic limitations. The grey region depicts the time beyond 4.5 solar h, wherein the device operation becomes impractical. **e**, Example device performance as a result of packing porosity ε. The total amount of water sorption linearly decreases with ε, whereas the intercrystalline diffusivity can be largely improved, whose limits are set by vapour diffusivity in random pores and the straight channel. The grey band shows the optimal regime with a sufficient amount of water sorption and fast kinetics.

Box 1

Definitions of variables

C_m: total system cost per unit area, US\$ m^{-2}
C: vapour concentration, mol m^{-3}
C_{eq}: equilibrium concentration of vapour from the sorption isotherm, mol m^{-3}
C_μ: vapour concentration within the sorbent material crystals, mol m^{-3}
c_p: specific heat capacity, J kg^{-1} K^{-1}
D_w: water diffusivity, m^2 s^{-1}
D_μ: intracrystalline diffusivity, m^2 s^{-1}
D_{vap}: vapour diffusivity in the local environment, m^2 s^{-1}
$D_{k,vap}$: Knudsen diffusivity of water vapour, m^2 s^{-1}
D_p: diffusivity in porous media, m^2 s^{-1}
ε: packing porosity of a porous medium, that is, the ratio of the void volume to the total volume of the medium
η_t: thermal efficiency of a system
h_s: enthalpy of sorption, kJ mol^{-1}
h_{fg}: heat of vaporization of water, kJ mol^{-1}
I_{irr}: daily solar irradiance, kWh m^{-2} day^{-1}
k: thermal conductivity, W m^{-1} K^{-1}
Λ: mean free path, nm
m_w: mass of harvested water, g
P_w: the unit price of water, US\$ l^{-1}
q_{max}: maximum water uptake of water sorption isotherms, g g^{-1}
Q: total amount of energy input in a device operation, J
ρ: density of materials, kg m^{-3}
r_c: crystal radius, μm
r_m: the size of a water molecule, nm
r_p: the pore size of nanoporous materials, nm
RH: relative humidity
RH_I: RH inflection point of water sorption isotherm wherein the step occurs
SWP: specific water productivity, l m^{-2} day^{-1}
SEC: specific energy consumption, kWh l^{-1}
τ_g: tortuosity factor of a porous medium
t: time, s
t_{life}: system lifetime, day
T: temperature, °C
T_{des}: desorption temperature, °C

volume. We note that water sorption kinetics of nanoporous materials and hydrogels are fundamentally different owing to their distinct water transport mechanisms. As a critical difference, hydrogels can accommodate liquid water transport – which is accompanied by a change in hydrogel morphology owing to swelling – in addition to vapour diffusion. The sorption kinetics of hydrogels for SAWH devices are less studied than those of nanoporous materials, which hinders both material-level and device-level optimization[39,45,131], although recent studies have shown that their sorption kinetics can be described with a similar TCM framework by incorporating the liquid water concentration of the hydrogel[39,131].

For devices using nanoporous materials as sorbents, water sorption kinetics depend on intracrystalline diffusion inside material crystals, intercrystalline diffusion within the voids between crystals in the packed sorbent layer (Fig. 4b), and vapour transport to the condenser, which correspond to distinct diffusion regimes at the different characteristic length scales or pore sizes r_p (Fig. 4c). Water diffusivity is determined by r_p, the mean free path of water molecules Λ and the size of the water molecule r_m, which can vary considerably[58]. Figure 4c highlights the range of diffusivity of surface diffusion ($r_p \approx r_m$), Knudsen diffusion ($r_p \approx \Lambda$) and molecular diffusion ($r_p \gg \Lambda$), corresponding to the engineering space for optimizing sorption kinetics at different length scales[58]. The intracrystalline diffusion is where the surface diffusion dominates as the pore size r_p is comparable to the size of the water molecules ($r_m \approx 3$ Å). The intracrystalline diffusivity D_μ can vary several orders of magnitude with a minor change of r_p because of strong nanoconfinement effects. The sorption kinetics within the crystals can be described using the linear driving force model[56]:

$$\frac{\partial C_\mu}{\partial t} = \frac{15}{r_c^2} D_\mu (C_{eq} - C_\mu), \quad (2)$$

where t is time, r_c is the radius of the material crystals and C_{eq} is the equilibrium concentration of vapour given by the sorption isotherm. In the sorbent layer, the intercrystalline diffusion includes both Knudsen diffusion when the size of voids (r_p of the packing) between crystals is comparable to Λ, and molecular diffusion when $r_p \gg \Lambda$ (Fig. 4b). Vapour transport to the condenser across the air gap, simply governed by molecular diffusion, is described by Fick's law of diffusion, $\frac{\partial C}{\partial t} = D_{vap}\nabla^2 C$, where D_{vap} is the vapour diffusivity in the ambient. Inside the packed sorbent layer, mass transport can be modelled as[9,10,23,37]

$$\frac{\partial C}{\partial t} - \nabla \cdot D_p \nabla C = -\frac{(1-\varepsilon)}{\varepsilon}\frac{\partial C_\mu}{\partial t}, \quad (3)$$

where D_p is the diffusivity in porous media. The kinetic limitations at different length scales introduce resistances to the sorption performance of material crystals, the sorbent layer and the device. Figure 4d shows the large differences in the evolution of water release from the material, the sorbent layer and the device during desorption, which are attributed to the resistances of intercrystalline diffusion and vapour transport to the condenser (Fig. 4b). The grey area in Fig. 4d marks the period beyond 4.5 solar h during which device operation becomes impractical. This underscores the critical need to maximize the use of solar energy within a limited period by minimizing kinetic constraints. The resistance of intercrystalline diffusivity dominates kinetic limitations in the current SAWH device design[23,25,37,132]. Therefore, one of the most critical gaps to be filled is through the improvement of intercrystalline diffusivity, whose upper and bottom limits are set by D_g and D_l, the diffusivity of pure water in the gas phase and liquid phase, respectively[58,59] (Fig. 4c). With consideration of both Knudsen diffusion and molecular diffusion in tortuous porous media, D_p can be estimated as

$$D_p = \frac{\varepsilon}{\tau_g}\left(\frac{1}{D_{vap}} + \frac{1}{D_{k,vap}}\right)^{-1}, \quad (4)$$

where τ_g is the tortuosity and $D_{k,vap}$ is the Knudsen diffusivity, of which the contribution can be neglected at moderate packing porosity[9,37]. For a randomly packed sorbent layer where $\tau_g = \varepsilon^{-1/2}$, $D_p = \varepsilon^{3/2}\left(\frac{1}{D_{vap}} + \frac{1}{D_{k,vap}}\right)^{-1}$. For a well-packed sorbent layer with perfectly aligned channels, $\tau_g = 1$, that is, the straight channel limit, $D_p = \varepsilon\left(\frac{1}{D_{vap}} + \frac{1}{D_{k,vap}}\right)^{-1}$. Therefore, a higher ε and unit tortuosity are expected to improve sorption kinetics. However, the desired ε is dictated by a design trade-off between the total amount of water harvesting and sorption kinetics. The total amount of water that can be captured by the sorbent layer decreases linearly with ε, and vapour diffusivity in the random packing and in the straight channels both monotonically increase with ε (Fig. 4e). A lower ε (or higher packing density) can substantially reduce mass transfer, especially when the crystal sizes are small, owing to Knudsen diffusion[9,37]. The opposite trend of water diffusivity and water sorption with ε shows that it is essential to carefully design ε to achieve the optimal device performance (Fig. 4e, grey band). The difference of D_p with the packing of random pores and straight channels also highlights the need to improve sorption kinetics through novel sorbent packing strategies[37,133,134].

Performance enhancement by thermal engineering

The coupling of water vapour transport with heat transfer governs the device performance, especially for desorption. The mechanistic framework of sorption kinetics can be integrated with the following equation to capture transient heat transfer in the sorbent layers[9,37]:

$$\rho c_p \frac{\partial T}{\partial t} - \nabla \cdot k \nabla T = h_s (1-\varepsilon) \frac{\partial C_\mu}{\partial t}, \tag{5}$$

where ρ, c_p and k are the effective density, specific heat capacity and thermal conductivity of the sorbent layer, respectively. The overall framework (equations (2)–(5)) is capable of predicting sorption kinetics and heat transfer performance of a SAWH device with macroscopic water sorption properties (Fig. 2c–e) as inputs, which serve as general guidelines on bridging materials innovations and device realization[9,37].

Beyond vapour transport within the sorbent layer, we discuss the coupled heat and mass transfer throughout the entire SAWH device involving multiple components. Figure 5a shows the overall picture of the coupled heat and mass transfer of a SAWH device. During device operation, the solar absorber and the sorbent layer can reach a high temperature for desorption. Heat can be carried through vapour diffusion and conduction across the air gap and released by condensation, developing temperature and vapour concentration gradients across the gap. To improve device performance, it is important to minimize heat loss and increase the desorption temperature T_{des} for more efficient desorption. With optimal thermal insulation, heat can be more efficiently delivered to overcome the energy barrier of h_s, leading to the enhancement of η_t. The main mechanisms of heat loss include convection and radiation from the top of the device and sidewall to the far field. Thus, the thermal performance can be improved through appropriate sidewall insulation and adding a transparent convective cover above the solar absorber. Vapour leakage is also detrimental to device performance and can be prevented with sufficient sealing. In addition, typical solar absorbers are not as black as ideal (≈90% solar absorptance), so implementing a darker surface (≈95% solar absorptance) can be an effective strategy to harvest more sunlight. A spectrally selective absorber with low IR emissivity or a silica aerogel cover can further reduce radiative heat loss and, hence, improve device performance[10,135]. It is worthwhile to note that the thermal performance does not always improve with a higher T_{des} because sensible heat and heat loss to the ambient may dominate[51,136,137].

Advanced device configurations

Innovations in device configuration are also key to optimizing sorption kinetics and heat transfer. So far, in addition to the above SAWH device design (Fig. 5a), only a few advanced configurations exist; we highlight three representative devices because of their different working principles. Inspired by multistage solar distillation, a dual-stage SAWH device was developed by recycling the latent heat of condensation from the top stage to assist in desorption of the bottom stage[23] (Fig. 5b). This configuration increases the SWP of a device operating on a daily cycle and can fundamentally enhance η_t. By desorbing two sorbent layers sequentially under a single solar absorber, that is, with the same device footprint, the design allows better utilization of the input energy. Solar-driven devices, however, can operate only in a diurnal operational mode, and their SWPs may be limited by sorption kinetics. Therefore, a cyclic SAWH device design with a fan for adsorption and a Joule heater for desorption was proposed, with electricity generated by PV cells[112] (Fig. 5c). The air flow can not only supply water vapour but can also cool down the heated sorbent layer after desorption. More recently, another cyclic water harvester was designed to operate in a batch-processed manner, wherein multiple saturated sorbent layers were desorbed alternately[27]. Using a high-performance LiCl-based hygroscopic composite as the sorbent, this device achieved an exceptional amount of water production[27].

Figure 5d compares the state-of-the-art solar-driven and PV cell-powered SAWH devices discussed above in terms of three performance metrics. For solar-driven devices, ① is a representative small-scale single-stage device prototype[10], whereas ② and ③ are relatively large-scale devices with single-stage and dual-stage configurations, respectively[23]. By contrast, PV cell-powered devices (④ (ref. 112) and ⑤ (ref. 27)) operate with multiple sorption–desorption cycles, and their η_t and SEC were calculated based on a solar-to-electricity conversion efficiency of 20%[24,134,138]. The large differences of η_t and SWP are because of the distinct mechanisms between devices, which also indicates that different SAWH devices should be designed with different optimization strategies. Among solar-driven devices, the η_t of a small-scale device ① is higher than those of large-scale devices ② and ③. This is because when a device is scaled up, heat loss and the challenge of water collection become more substantial, leading to a lower η_t. However, the SWPs of ② and ③ are higher than that of ①, as these large-scale devices were carefully optimized with respect to SWP and loaded with more sorbent materials per solar absorbing area[23]. From ② to ③, there is also a performance enhancement because the latent heat is reused in the dual-stage configuration[23]. Surprisingly, PV cell-powered devices do not demonstrate sizable performance advantages compared with solar-driven devices. In fact, although cyclic devices may have the potential to achieve high water production, they are limited by a low η_t and a high SEC owing to the low solar-to-electricity conversion efficiency of PV cells (≈20%)[24,134,138]. A large area of solar panels is required to supply sufficient electricity for activating device mechanisms, which further constrains SWPs[134]. In addition to energy consumption, considerations regarding the time period of device operation cycles differ for a daily cycle device and a cyclic device, and can be more important to the latter. The cyclic usage of auxiliary components in ④, such as heaters and fans, may lead to complex coupling between sorption kinetics and heat transfer, which requires more in-depth analysis and careful optimization[112]. Even though ⑤ uses a novel sorbent with a high water uptake, the SWP is not substantially higher than those of other devices[27]. This suggests that there may be a gap between materials innovations and device realization in PV cell-powered devices that hinders their optimization.

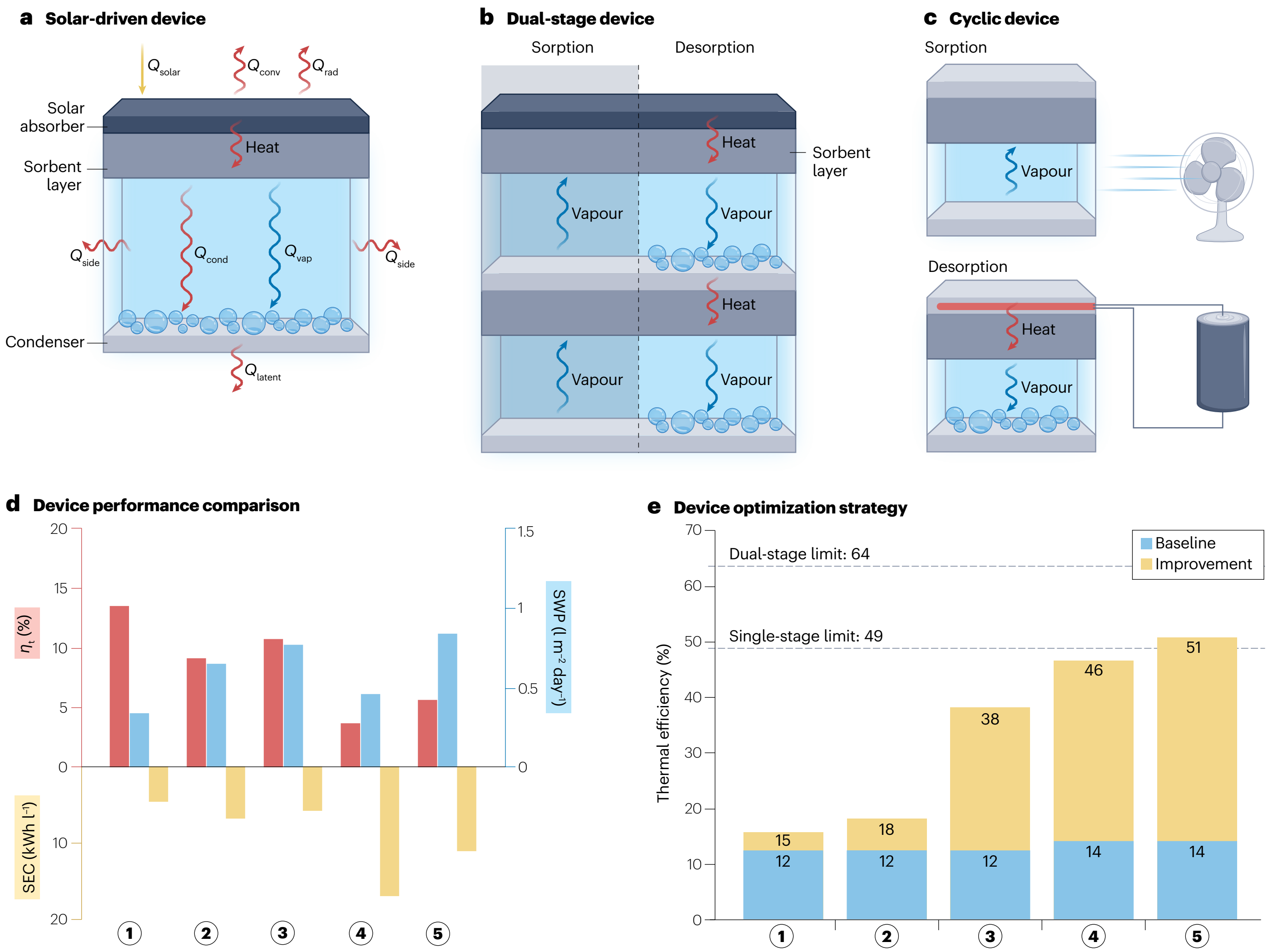

Fig. 5 | Advanced design strategies and opportunity space to reach thermodynamic limits. a, Heat flow diagram of a solar-driven sorption-based atmospheric water harvesting (SAWH) device. Sunlight is converted to heat by the solar absorber and then conducted through the sorbent layer to trigger desorption. Heat can be further carried through vapour diffusion and conduction across the air gap, and released through condensation. Temperature and vapour concentration gradients will be built up across the air gap. Heat loss arises from convection to the ambient, radiation from the top of the device, and imperfect sidewall insulation. **b**, A dual-stage SAWH device. The heat released by condensation on the top stage condenser is recycled to drive the desorption of the bottom stage sorbent layer. **c**, A cyclic SAWH device, typically powered by solar cells. Energy is consumed for air supply during sorption and for Joule heating during desorption. **d**, Performance comparison of the state-of-the-art systems in terms of solar-to-water thermal efficiency η_t, specific water productivity (SWP) and specific energy consumption (SEC). Solar-driven, daily cycle devices: ① (ref. 10), ② (ref. 23), ③ (ref. 23); photovoltaic-powered, cyclic device: ④ (ref. 112), ⑤ (ref. 27). **e**, Improving the thermal efficiency of a zeolite-based SAWH system with different optimization strategies towards the thermodynamic limits (dashed lines). Blue areas represent the baseline thermal efficiency of single-stage and dual-stage devices without optimization. Yellow areas represent thermal efficiency improvement owing to ① better sidewall insulation, ② a darker solar absorber, ③ the use of a transparent convection cover in a single-stage configuration, ④ a dual-stage architecture with the aforementioned optimization strategies, and ⑤ optimized dual-stage architecture with optimal vapour gap thicknesses. Considerable improvement is achieved when each optimization strategy is added. With careful design, the dual-stage system can surpass the thermodynamics limit of a single-stage device. cond, condensation; conv, convection; rad, radiation; vap, vaporization.

Towards thermodynamic limits

The insights gained from device optimization shed light on the large opportunity space to further improve SAWH devices towards their thermodynamic limits. As an example, we provide a quantitative thermal analysis of SAWH devices using the zeolite AQSOA Z01, which is a commercial sorbent material used for SAWH applications[23]. Using a recently developed thermodynamic model[136], we predict that the energy efficiency limits of a single-stage device and a dual-stage device are 49% and 64%, respectively. Figure 5e shows the engineering space and several design strategies to reach thermodynamic limits. Blue areas represent the baseline η_t of single-stage and dual-stage devices without optimization (12% and 14%, respectively), whereas yellow areas represent the improvement of η_t through different strategies (Fig. 5e). The η_t was evaluated based on equations (1)–(5) and desorption was assumed to be triggered by one-sun illumination lasting for 4.5 h, which is the average daily solar irradiance of the USA[139]. The η_t of a single-stage device can be enhanced via the minimization of heat loss through conduction through sidewall (①), through convection to the ambient (②) and by harvesting more sunlight (③). Each of these mechanisms contributes substantially to the overall enhancement of device performance compared with the baseline device performance (Fig. 5e). The η_t can be further increased in a dual-stage device (Fig. 5b). By optimizing heat transfer using these three mechanisms (①–③), the η_t of a dual-stage device can be enhanced from 14% to 46% (④). The air gap thickness also has an important role in the dual-stage device optimization because it determines the temperature and vapour concentration gradients from the solar absorber to the condenser, wherein an optimal geometry exists (⑤)[23]. The η_t of a dual-stage device with an optimal air gap thickness can reach 51%, surpassing the thermodynamic efficiency of a single-stage device (49%, Fig. 5e).

Innovative approaches are desired to further enhance sorption kinetics, such as novel packing solutions of sorbents[37,133,134] and engineering swelling kinetics of hydrogels[39,42], and to improve the thermal performance via passive cooling[10,140,141], concentrated solar power[4,10] and the suppression of IR emission using aerogels[10,135]. With sufficient heat concentration and localization, a multistage SAWH device can be achieved with further performance enhancement[23,142,143]. Because of the constraints of sunlight, the exploration of device mechanisms using other energy sources, such as low-grade waste heat, is valuable as well[53,131,134].

Real-world impact of SAWH technologies

Although there are clear pathways to further promote SAWH technologies through the development of materials and devices, the real-world impact of large-scale SAWH systems on the water economy, society and the environment is yet to unfold. Unlike conventional AWH technologies, SAWH has the universal capability to operate using low-grade energy and has no apparent geospatial limitation[9,21,24]. Scaling SAWH from the laboratory into practice can, thus, be a futuristic goal aimed at the global scale[24]. Herein, we point out several roadblocks and opportunities for deploying SAWH systems in practice by examining the technoeconomic feasibility, and we then provide an outlook of SAWH technologies on society and the environment at the global scale.

System developments and opportunities

Scaling up of devices to satisfy the water demand of a household and a community reliably requires development at the system level. In addition to being highly water productive, SAWH systems are expected to have both materials and devices that are effective in the long term. Therefore, further efforts to integrate and deploy systems in a practical manner are required, such as air or water filtration, device maintenance, water storage and distribution, and potentially electronic control mechanisms. Different from other length scales, system-level developments of SAWH also require the more realistic considerations of cost and benefit, which is the gap between academic research and real-world applications. In fact, companies have already made SAWH systems commercially available. We envision that as commercial products develop, more insights can be gained and shared among academia, companies and consumers to better understand the requirements at the system-level. Here, we identify a few challenges of SAWH systems that remain to be addressed by research endeavours.

First, sorbent materials are required to be bonded onto other device components, such as a thermal binder. The stability of this attachment determines the long-term performance of the sorbent and the device lifetime. For example, poor coating of nanoporous materials onto a porous metal foam binder can block the pores of the sorbent layers and slow down sorption kinetics[37,130]. Hydrogels may need to be covalently bonded to the substrate to ensure good thermal contact[49]. The performance and robustness of the condenser is another critical challenge that has been largely overlooked[144]. When exposed to the ambient, the condenser surface degrades over time owing to hydrocarbon adsorption and sorbent particle contamination, which leads to considerable water retention, making it a bottleneck of water collection[23]. An ideal condenser surface should be chemically stable and can facilitate water collection and reject heat efficiently. Therefore, a condensing surface is expected to be either highly hydrophilic or superhydrophobic with minimal contact angle hysteresis, which can be realized through structured surface fabrication and coating treatments[23,144].

Another concern that remains to be addressed is the quality of water produced from SAWH[145,146]. Possible contamination may come from several sources, such as aluminium[23], iron[10,23], chromium[9] and chlorine[48,117] elements from sorbent materials, and copper[9,23] and nickel[23] elements from metallic device components. A 2021 study found that the concentrations of aluminium, iron and silver ions in collected water from a SAWH device were above the Environmental Protection Agency secondary drinking water standard[23]. The concentrations of nickel and copper ions were substantial too[23]. In addition, an excessive amount of Li^+ in the water, resulting from wide use of the moderately toxic LiCl as a sorbent, can be harmful for human health[48,117]. Aside from inorganic contaminants, waterborne pathogens and organic pollutants may need to be removed from the water before it is delivered for drinking[7,145]. Air pollution may also affect decisions of whether to adopt SAWH, which still demands more investigations[145]. Strategies for minimizing contamination and purification must be considered during system-level development[7].

Technoeconomic feasibility of SAWH

The technoeconomic feasibility of SAWH technologies has been one of the main driving forces for system-level development to enable real-world impact on the water economy. Currently, access to drinking water from the point of production to the end of use requires the transportation, distribution, sanitation and hygiene of water, and the capitals of infrastructures and maintenance, all of which can have a considerable impact on the cost of water production. By contrast, the total cost of SAWH includes only the costs of materials, device manufacturing, system deployment and maintenance. Therefore, with careful design and proper optimization, installing SAWH systems to provide drinking water can be a cost-effective solution, especially in arid or low-income areas.

To understand the commercial potential and challenges of SAWH technologies, we perform a technoeconomic analysis of solar-driven SAWH systems. For a fair and consistent comparison, we consider only the costs of materials and device components, whereas the manufacturing cost is not included. The operational cost is neglected for simplicity because of the passive nature of these systems. Figure 6a shows the unit price of water P_w (US\$ l^{-1}) as a function of lifetime, which can be estimated by

$$P_w = \frac{C_m}{t_{life} I_{irr} \frac{1}{SEC}}, \quad (6)$$

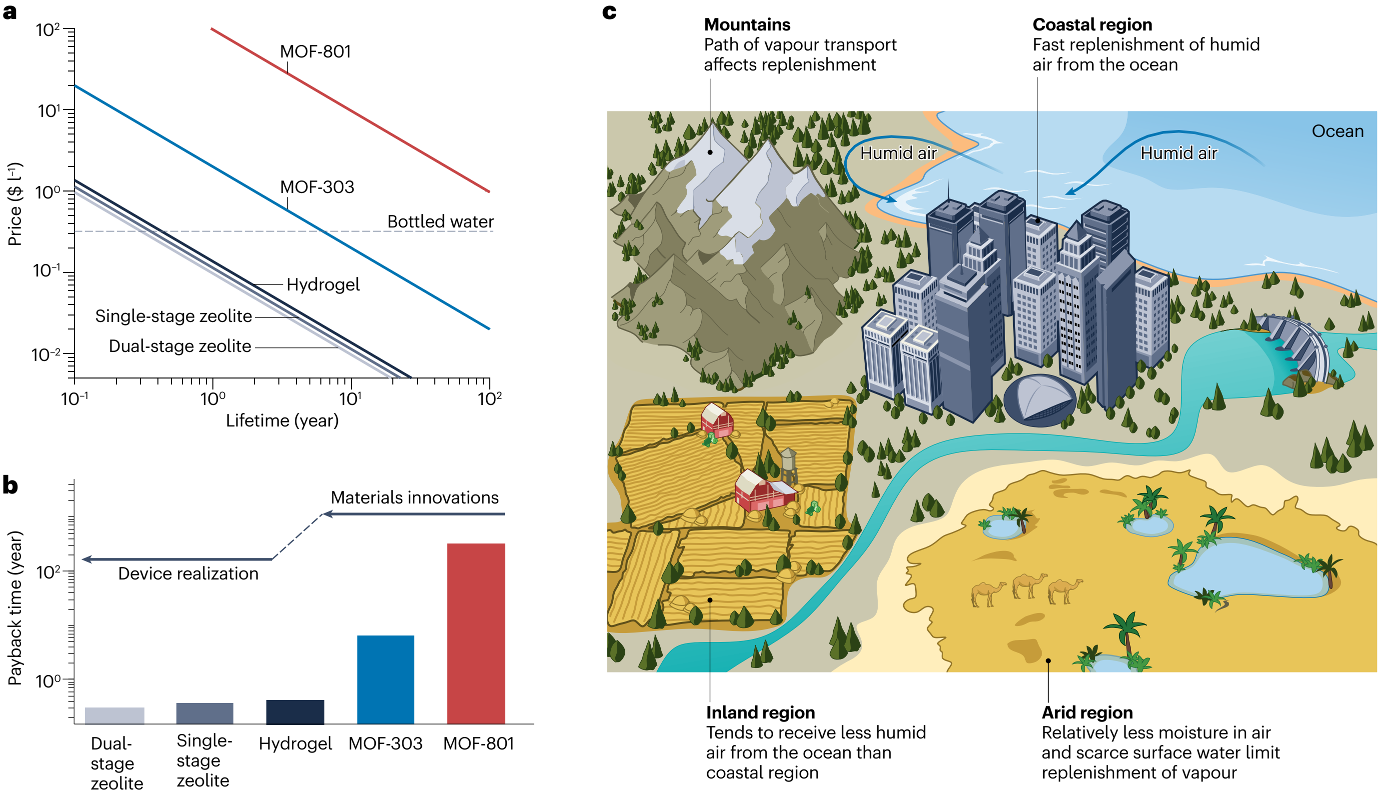

Fig. 6 | Real-world impact on the worldwide water economy and environment. a, Price of sorption-based atmospheric water harvesting (SAWH) water as a function of the system lifetime compared with the bottled water price of 0.335 US\$ l^{-1} (dashed line). This analysis is based on the total cost of the sorbent materials and device components with different configurations. **b**, Payback times of SAWH technologies when the price of SAWH-produced water is equal to the bottled water price. The technoeconomic analysis shows the potential of commercializing SAWH technologies and elucidates the synergy of materials innovations and device realization towards real-world impact. **c**, Water sorption in the hydrological cycle on Earth and its geospatial variations. When a substantial amount of water vapour is extracted from the atmosphere, the large-scale SAWH could alter the equilibrium of the climate system and intervene in the hydrological cycle. The redistribution of water vapour through diffusion and advection is induced to replenish extracted water vapour. Coastal cities can receive more and fast replenishment of humid air from the ocean. The replenishment of vapour will be more critical to inland areas and arid regions, where surface water is extremely limited, and can be hindered by mountains. Consequently, local climates of inland areas and arid regions might be more sensitive to substantial extraction of vapour from air. MOF, metal–organic framework.

where C_m is the total system cost per unit area (US\$ m^{-2}), t_{life} is the system lifetime (day), and daily solar irradiance I_{irr} = 4.5 kWh m^{-2} day^{-1} was used in the calculation[139]. Therefore, SAWH will be more economically feasible in regions with higher solar irradiance. The analysis shows that SAWH water derived using low-cost, scalable sorbent materials promises to compete with the bottled water price of 0.335 \$ l^{-1} (ref. 5) (Fig. 6a, dashed line). Taking the single-stage device with zeolite AQSOA Z01 sorbent as an example[23] (Fig. 6a, orange solid line), the total capital cost per unit area is 144.9 \$ m^{-2}. With a maximum thermal efficiency of 49%, the ideal SEC of a passive SAWH system is 1.28 kWh l^{-1}. Consequently, the payback time of the single-stage SAWH device using AQSOA Z01, wherein the price is equal to the cost of bottled water, is approximately 4 months (Fig. 6b). When the lifetime of a SAWH system is larger than the payback time, it becomes economically feasible. From a practical consideration, the payback period should be within a few years. In the near future, achieving a lower cost and improved scalability will be crucial for MOFs to outperform other sorbents[118,120,121]. For example, in a single-stage device based on MOF-801, a costly sorbent used in the demonstration of SAWH in 2017 (ref. 9), the payback time is estimated to be hundreds of years (Fig. 6b). When a more scalable MOF-303 is used for a single-stage device[11], the payback period can be shortened to realistically 6 years (Fig. 6b). Although hydrogels, such as polyacrylamide–carbon nanotube–$CaCl_2$ used in our analysis[26], typically have a much higher water uptake than zeolites, the payback time can be longer. This is because their thermodynamic efficiency and material cost may not be competitive enough in certain conditions, especially in an arid environment. However, it is crucial to acknowledge that the costs of sorbent materials in the analysis can change over time. We envision that as MOFs are scaled up more efficiently and cost-effectively[147], and as high-performance hydrogels are demonstrated and commercialized[38,42,45], their technoeconomic performances tend to substantially improve.

When the materials cost becomes a smaller fraction of the total cost, device and system optimization becomes critical to making SAWH more economically feasible in terms of energy efficiency and lifetime. For example, the payback time is estimated to be approximately

3.5 months when using a dual-stage rather than a single-stage architecture in a AQSOA Z01 device (Fig. 6b). The reduction of payback time is attributed to the thermal efficiency enhancement, although it is also compensated by additional materials and device costs. In our analysis, the cost of each device component is estimated based on the retailer prices, which can be considerably reduced with a well-established manufacturing and assembly line, leading to further shortened payback times. In addition, increasing the system lifetime can also reduce the water price, which suggests the importance of having a reliable system with a long lifetime. The technoeconomic analysis clearly shows the pathways to achieve the real-world impact of SAWH through the multiscale development of materials, devices and systems.

Humanitarian, societal and development impact

The SAWH technologies also have tremendous positive impact on water, sanitation and hygiene[1,3]. In regions with no or limited infrastructures for drinking water, going out to find water resources is one of the primary manners to obtain drinking water, which raises concern about water quality, results in lost work or study chances, and amplifies inequality[1]. Installing off-the-grid SAWH systems will provide affordable infrastructures in these areas to secure access to clean water and avoid water transportation through the long-distance pipeline. It also has the potential to address urgent water needs owing to natural disasters, quarantine or other humanitarian emergencies[1,3,148,149]. Therefore, developing SAWH to create a clean water source is societally meaningful to empower communities and decrease premature deaths[1,8]. The point of access to high-purity water enabled by SAWH technologies requires zero or minimum dependence on an electrical grid to produce water and may eliminate or largely alleviate the need for water transportation and maintenance. Therefore, other than providing drinking water, SAWH also holds promise to supply clean water cost-effectively for pharmaceutical and industrial applications that require pure water[2], and to provide cooling power for space conditioning[18–20] and electronic thermal management[54,150].

Outlook

This Review provides a multiscale perspective for SAWH technologies to guide their next-generation designs and facilitate their broader impact on society and the environment. Still, a large opportunity space of SAWH technologies remains to be explored. More physical insights can be expected from accurate characterization and molecular simulations of complex sorbent systems. The combined efforts of experiments and modelling promise a mechanistic understanding of how microscopic sorption mechanisms dictate macroscopic water sorption properties. Innovative solutions to further enhance sorption kinetics and heat transfer will be essential. The stability and durability of materials and device components require more investigation to prolong the lifetime of SAWH systems. We anticipate that a complete multiscale design strategy from the molecular to the device level can achieve the optimal performance of next-generation SAWH devices. With guidance from the device and thermodynamic models based on climatic conditions, optimal macroscopic water sorption properties and device design parameters can be determined. With the molecular-level understanding of water sorption mechanisms enabled by advanced metrology and molecular simulations, one can engineer chemical and structural features of materials to achieve desired water sorption properties. The synergy of design at different length scales can, thus, bridge materials innovations to device realization and push the performance of SAWH approaching the thermodynamic limits.

Although a bright future of SAWH can be envisioned, little is known about its scaled-up impact on the environment. When a substantial amount of moisture is extracted from the atmosphere to meet the water demands of billions of people, there may be a notable impact on the natural hydrological cycle of Earth. A typical hydrological cycle involves evaporation, condensation, precipitation and percolation[151–154]. When a considerable amount of water vapour is extracted, SAWH can intervene in condensation and precipitation processes that might alter the equilibrium of the climate[24,152]. For this reason, the redistribution of water vapour through diffusion and advection is induced at the geological time and length scales[153,155]. Water vapour may diffuse, from the outer atmosphere or farther regions, to replenish extracted water vapour. The timescale of this replenishment process is critical during the implementation of SAWH (Fig. 6c). Because of high evaporation in the ocean, coastal areas receive more humid air and typically have higher RH and more precipitation than inland areas[153,155]. The transport of vapour replenishment will, thus, be more critical to inland areas, especially arid regions, where surface water is extremely limited. Meanwhile, the path of transport may also affect the replenishment; for example, mountains may hinder humid-air transport[153,155]. Therefore, the global implementation of SAWH will need to take into account the interplay between transport of replenishment and evaporation in the waterbodies and surface water[151,153–155].

More importantly, the sizable extraction of water vapour from the atmosphere — and the ability to regulate moisture in the atmosphere — potentially provides a new perspective on the greenhouse effect. It is a common belief that capturing non-condensable gas in the atmosphere, that is, carbon dioxide, and reducing carbon footprints are the most effective approaches to mitigating climate change[156–158]. However, this conclusion was drawn from an analysis of non-condensable gas assuming that there are no feasible ways to actively intervene in the hydrological cycle to drive the motion of water vapour[159]. In fact, water vapour is the most abundant greenhouse gas and contributes to about 50% of the greenhouse effect of Earth, with CO_2 contributing only 20%[156]. Although water vapour is not perceived as the driver of global warming, it forms a positive feedback relation with the global temperature and amplifies global warming[157,159,160]. A global-scale SAWH may affect cloud formation, precipitation and geospatial RH distribution, which may alter the radiation balance of Earth[152,158–160]. Therefore, the global potential of SAWH suggests the possibility of mitigating the greenhouse effect. The actual impact of scaling SAWH on the positive feedback loop of water vapour in the radiation balance of Earth remains to be revealed. We raise this open question to be addressed by combining research into SAWH, climate science and geoengineering. To develop next-generation SAWH technologies that are capable of addressing global challenges of water scarcity and climate change, it will be essential to have input from chemists, material scientists, environmental scientists and engineers.

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Y.Z. and L.Z. contributed equally to this work. Y.Z., L.Z. and E.N.W. conceptualized the manuscript. Y.Z., L.Z., X.L., B.E.F. and C.D.D.-M. researched data and performed analysis for the article. L.Z. and Y.Z. conceived and illustrated the figures and tables. All authors contributed to the discussion of content, writing, and editing of the manuscript before submission.

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